Finding Resonance in Speech and Song

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LabPhon 20, Montréal, 27 June 2026

This poster involves re-analysis of data from my 2010 MPhil. Big thanks to:



Allan Bell
(AUT/Laidlaw)



Jen Hay
(U. Canterbury)

Some shifts to Pop Song English (PSE) are solely dialectal,
but opener vowels on high f0s (Austin 2007) may reinforce dialect-based shifts.

Formant-tuning is *not* a general singing technique for these singers.

Pop Song English

- American-influenced PSE: a supralocal norm in pop music (Gibson 2024)
- Early formation through intentional acts of identity (Trudgill 1983)
- Continues to dominate via context-dependent associative learning (Gibson & Hay 2025)

Research gap:
Singing technique as additional explanation for singing accents?

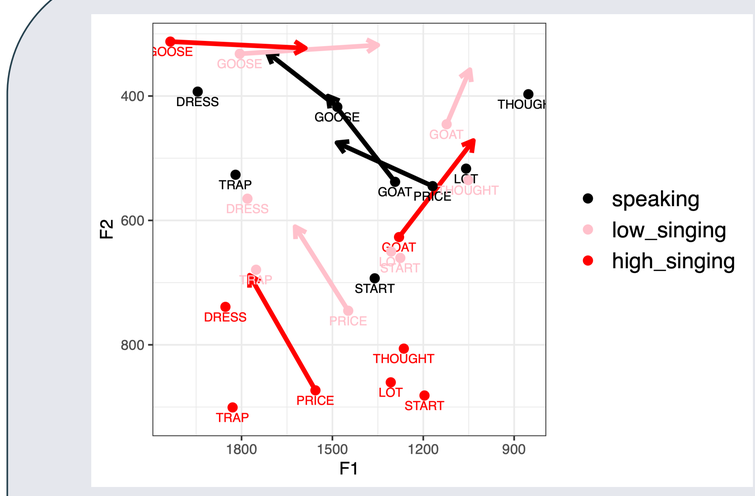


Figure 1. Mean vowel formants across 3 singers.
High singing (red), low singing (pink), speech (black)
"High singing": f0 > singer's max spoken f0. Labels at nucleus.

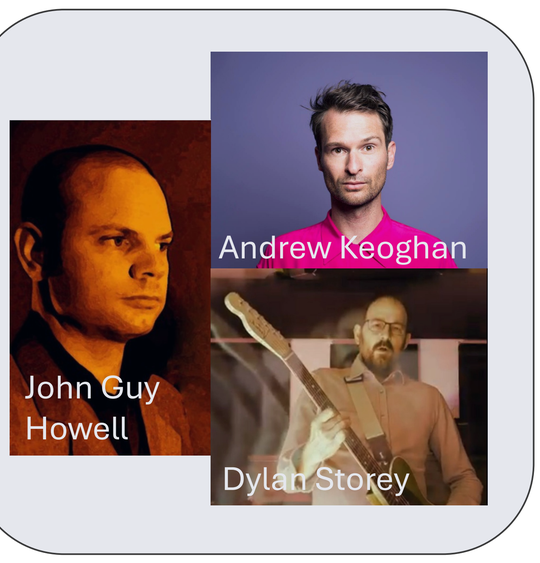
Singing technique effect on F1: high singing > low singing > speech
PSE dialect features: no diphthong shift; /uː/ & /əʊ/ trajectories

Formant-tuning

Alignment of formants with harmonics:

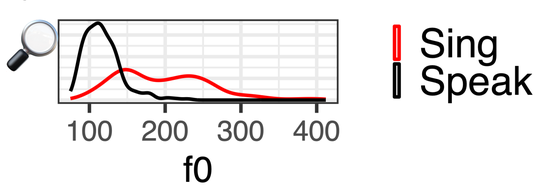
- Sopranos keep F1>f0 (Joliveau et al. 2004), musical theatre singers align F1 with 2*f0 for belt (Herbst et al. 2023)
- Alignment of F1 to n*f0 in clear speech (Story et al. 2025)
- Occasional tuning of F1/F2 'by chance', rather than a general singing strategy, in professional male singers (Sundberg et al. 2013: 287)

RQ: Does formant-tuning affect vowel choices for male New Zealand singer-songwriters?



Methods

- 3 New Zealand male singer-songwriters (data from Gibson 2010; Gibson & Bell 2012)
- Sung/recited lyrics + interview (high_sing n=107; low_sing n=203; recite n=282; interview n=173)
- 8 vowels (Nucleus and offglide targets selected by hand in Praat; Measures of f0, F1/F2, duration, intensity)
- Formant-tuning analysis based only on nucleus of tokens with stable f0 (n=697)
- Distance to nearest harmonic (in Hz) calculated for F1 and F2
- Lesser **formant-harmonic distance (FHD)** out of F1 & F2 for modelling (Sundberg 2013)
- Problem: distinct f0 distributions; singing higher than speech = wider spaced harmonics



Results

- Fig 2: Raw results for F1 by f0. No overall evidence of greater formant-tuning in singing than speech.
- Final model restricted to f0 range min-sung-f0 ↔️ max-speech-f0, by-singer (Andrew sing n=14 😜, he sings higher than speech!)
- LMER model: *FHD ~ f0 * singer + mode + duration + (1|word)*
  - f0*singer: **FHD increases with f0** for all singers, but to a different extent
  - mode (sing vs. speak): **FHD is *higher* in singing than speech**, when other effects held constant
  - duration: **FHD is *lower* on longer vowels**
- Collinearity of mode and f0 make modelling difficult
- Fig. 3: Raw FHD with loess smooth suggests Andrew and Dylan may both do some formant-tuning in their higher registers (flatter smooths), while John does not
- *But note:* Instability of LPC formant estimates at high f0 due to widely spaced harmonics
  '*Harmonic attraction*' (Whalen et al. 2022; see also Story et al. 2025)

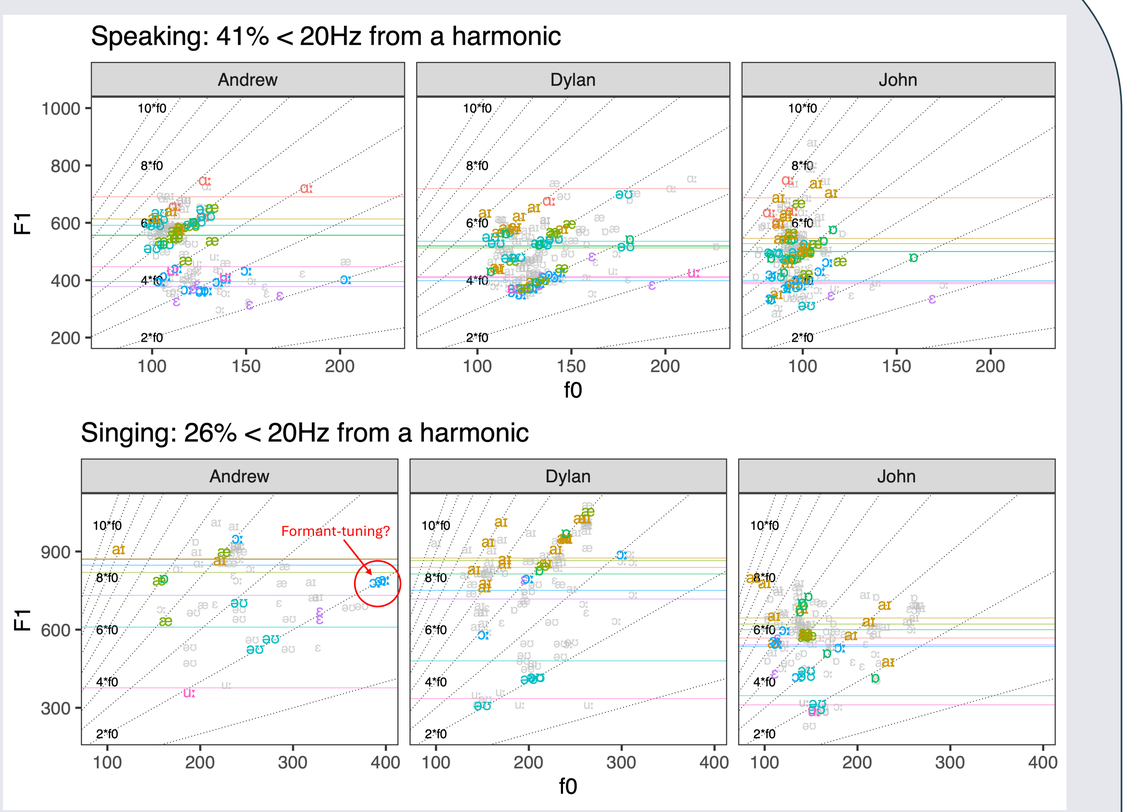


Figure 2. F1 and f0 . Diagonal lines show multiples of f0.
Horizontal line shows mean F1 per vowel, within singing/speaking.
If either formant < 20Hz from a harmonic, vowel point is coloured, otherwise grey.

Conclusions

- No strong evidence for systematic formant-tuning in these NZ males' singing as compared to speech
- Duration results suggest slightly closer formant-tuning in *both* singing and speech in longer vowels
- Opportunistic formant-tuning for certain vowel-f0 combinations

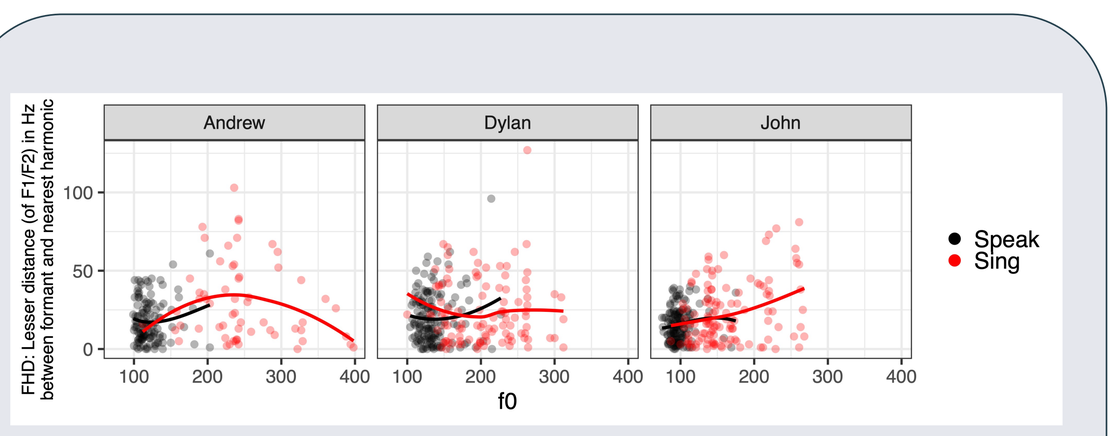


Figure 3. Formant-harmonic distance (lesser distance out of F1 & F2 to nearest harmonic) for sung (red) and spoken (black) vowels according to f0, with loess smooths (span = 1.2)

References **Austin (2007).** Jaw Opening in Novice and Experienced Classically Trained Singers. J. Voice 21(1) ||| **Gibson (2024).** Pop Song English as a supralocal norm. *Lang. in Soc.* 53(3). ||| **Gibson & Bell (2012).** Popular music singing as referee design. In Hernández-Campoy & Cutillas-Espinosa (eds.), *Style-Shifting in Public*. ||| **Gibson & Hay (2025).** Perceptual style-shifting across singing and speech: Music activates Pop Song English for NZ listeners. *Lang. & Speech*. ||| **Herbst et al. (2023).** Acoustical theory of vowel modification strategies in belting. *J. Voice* 39(5). ||| **Joliveau et al. (2004).** Vocal tract resonances in singing: The soprano voice. *JASA* 116(4). ||| **Story et al. (2025).** Detection of synchronization of the voice source and vocal tract in connected speech. *JASA* 158(3). ||| **Sundberg et al. (2013).** Formant tuning strategies in professional male opera singers. *J. Voice* 27(3). ||| **Trudgill (1983).** *On Dialect: Social and Geographical Perspectives*. ||| **Whalen et al. (2022).** Formants are easy to measure; resonances, not so much: Lessons from Klatt (1986). *JASA* 152(2).